Module - 3

Relations and Functions

Let A and B be 2 sets. Then the set of all ordered pairs (a,b) where $a \in A$ and $b \in B$ is called the cartesian product or cross product.

It is denoted by $A \times B$, Thus

$$A \times B = \{(a,b) / a \in A \text{ and } b \in B\}$$

Eg:- $A = \{-1, 0, 1\}$ and $B = \{2, 3\}$

$A \times B = \{(-1,2)\ (-1,3)\ (0,2)\ (0,3)\ (1,2)\ (1,3)\}$

$B \times A = \{(2,-1)\ (2,0)\ (2,1)\ (3,-1)\ (3,0)\ (3,1)\}$

$\therefore\ A \times B \neq B \times A$

$A \times A = A^2 = \{(-1,-1)\ (-1,0)\ (-1,1)\ (0,-1)\ (0,0)\ (0,1)\ (1,-1)\ (1,0)\ (1,1)\}$.

Note:- If $A = \{m\}$, $B = \{n\}$ then $A \times B = m \times n$ elemen

problems

(1) Let $A = \{1, 3, 5\}$, $B = \{2, 3\}$ and $C = \{4, 6\}$ find (i) $B \times C$ (ii) $(A \cup B) \times C$ (iii) $(B \times A) \cup (B \times C)$

soln:- (i) $B \times C = \{(2,4)\ (2,6)\ (3,4)\ (3,6)\}$.

(ii) $(A \cup B) \times C = \{1, 2, 3, 5\} \times \{4, 6\}$

VTUADDA

$= \{(1,4)\ (1,6)\ (2,4)\ (2,6)\ (3,4)\ (3,6)\ (5,4)\ (5,6)\}$

(iii) $(B \times A) \cup (B \times C)$

$(B \times A) = \{(2,1)\ (2,3)\ (2,5)\ (3,1)\ (3,3)\ (3,5)\}$

$(B \times C) = \{(2,4)\ (2,6)\ (3,4)\ (3,6)\}$

$\therefore (B \times A) \cup (B \times C) = \{(2,1)\ (2,3)\ (2,5)\ (3,1)\ (3,3)\ (3,5)\ (2,4)\ (2,6)\ (3,4)\ (3,6)\}$.

(2) Find x and y in each of the following cases

(i) $(2x, x+y) = (6, 1)$

(ii) $(y-2, 2x+1) = (x-1, y+2)$

Soln:- (i) $(2x, x+y) = (6,1)$

$2x = 6$ $\qquad$ $x+y = 1$

$\boxed{x = 3}$ $\qquad$ $3+y = 1$

$\boxed{y = -2}$

(ii) $(y-2, 2x+1) = (x-1, y+2)$

:- $y-2 = x-1$, $2x+1 = y+2$

$x-y+1 = 0$ $\qquad$ $2x-y-1 = 0$

$x - y + 1 = 0$

$2x - y - 1 = 0$

$-x + 2 = 0$

$\boxed{x = 2}$

$2(2) - y - 1 = 0$

$-y = -3$

$\boxed{y = 3}$

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(3) Suppose $A, B, C \subseteq Z \times Z$ where

$A = \{(x,y) / y = 5x - 1\}$

$B = \{(x,y) / y = 6x\}$

$C = \{(x,y) / 3x - y = -7\}$, Find (i) $\overline{\bar{A} \cup \bar{C}}$

(ii) $\overline{\bar{B} \cup \bar{C}}$

Soln:- (i) $\overline{\bar{A} \cup \bar{C}} = \bar{\bar{A}} \cap \bar{\bar{C}}$ $\left[\text{WKT } \overline{A \cup B} = \bar{A} \cap \bar{B}\right]$

$= A \cap C$ $\quad (A')' = A.$

$\Rightarrow (x,y) \in A \cap C \Leftrightarrow (x,y) \in A$ & $(x,y) \in C$

$\Leftrightarrow y = 5x - 1$ and $3x - y = -7$

$5x - y = 1$ $\quad 3x - y = -7$

$\therefore 5x - y = 1$

$(-)3x (-) y = (-) 7$

$2x = 8$

$\boxed{x = 4}$

$\therefore 12 - y = -7$

$-y = -19$

$\boxed{y = 19}$

(ii) $\overline{\bar{B} \cup \bar{C}} = \overline{B \cap C}$

To find, $B \cap C \Rightarrow y = 6x$ and $3x - y = -7$

$3x - 6x = -7$

$y = 14$ // $\quad -3x = -7$

$x = \frac{7}{3}$ //

$\therefore \left(\frac{7}{3}, 14\right)$

which is not possible $\because x = 7/3 \notin Z$

$\therefore B \cap C = \phi$ (null set)

VTUADDA

Since $B \cap C = \phi$

$\therefore \overline{B \cap C} = \overline{\phi}$ = universal set

$= Z \times Z$ //

(*) Relations

Defn: Let A and B be any 2 sets then a subset of $A \times B$ is called as the relation from A to B

i.e $R = \{(a,b) / a \in A, b \in B\}$

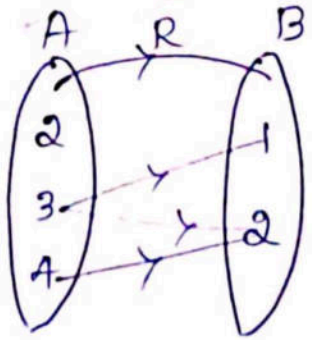
Eg:- $A = \{2,3,4\}$, $B = \{1,2\}$

$A \times B = \{(2,1)(2,2)(3,1)(3,2)(4,1)(4,2)\}$

$R = \{(3,1)(3,2)(4,2)\}$ $(R \subseteq A \times B)$

(or) $R = \{3R1, 3R2, 4R2\}$

Relation diagram $\Rightarrow$



It can be represented in diagram called as arrow diagram.

Note:- A contains m elements, B contains n elements then relation from A to B contains 2^{mn} elements.

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problems

(1) $A = \{1, 2, 3\}$, $B = \{2, 4, 5\}$ find the following

(i) $|A \times B|$? contains how

(ii) Number of relations from A to B.

(iii) No of binary relations on A

(iv) No of relations from A to B that contain (1, 2) and (1, 5)

(v) No. of relations from A to B that contain exactly 5 ordered pairs.

(vi) No. of relations from A to B that contains atleast 7 ordered pairs.

Soln:- (i) $|A \times B| = m \times n = 3 \times 3 = 9$

(ii) $A R B = 2^{mn} = 2^9 = 512$

(iii) No. of relation on $A = 2^{mn} = 2^{3 \times 3} = 512$

(iv) Let $R_1 = \{(1,2)(1,5)\} = 2$

$\therefore$ We've $|A \times B| = 9$

$\therefore$ no of relation is $= 9 - 2 = 7$

i.e $2^7 = 128$, no of relation from A to B contain the element (1,2) & (1,5)

(v) A R B containing only 5 ordered pairs are,

$^9C_5 = 126$ elements.

(vi) Atleast 7 ordered pairs.

possibilities are, $^9C_7 + {}^9C_8 + {}^9C_9 = 36 + 9 + 1 = 46$

VTUADDA

(2) Let $A = \{1, 2, 3, 4, 6\}$ and R be the relation on A defined as $(a, b) \in R$ if and only if a is multiple of b. Write down R as a set of ordered pairs.

Soln:- a is multiple of b

$$R = \{(a, b) / a, b \in A \text{ and 'a' is a multiple of } b\}$$

$$= \{(1,1)(2,1)(2,2)(3,1)(3,3)(4,1)(4,4)(4,2)(6,1)(6,2)(6,3)(6,6)\}.$$

(*) Functions :-

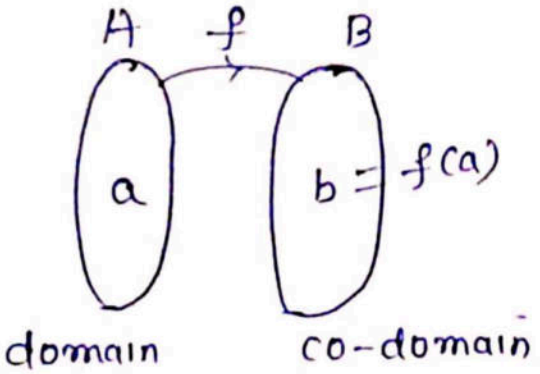
Let A and B be the 2 non-empty sets. Then A function or mapping 'f' from A to B is a relation from A to B such that every element of A has a unique element in B. i.e $b = f(a)$

Here 'b' is called the image of 'a' and 'a' is called the pre-image of 'b'

A function f from A to B is denoted by

$f : A \rightarrow B$

The pictorial representation of f is



$b = f(a) \rightarrow$ range of f.

VTUADDA

Note:-

(1) Every function is a relation but every relation need not be a function

(2) A contains m elements, B contains n elements then a function 'f' from A to B contains n^m elements.

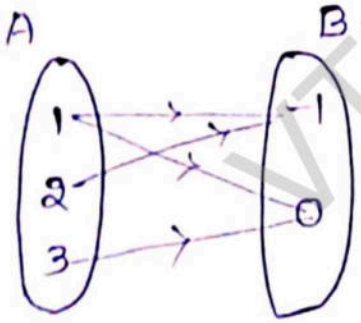
problems

(1) Determine the following relations are functions (or) not?

(i) $A = \{1, 2, 3\}$, $B = \{-1, 0\}$

$R = \{(1,-1)\ (1,0)\ (2,-1)\ (3,0)\}$

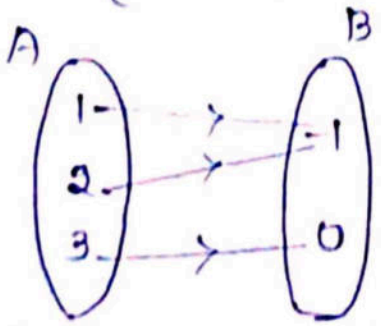
soln:- Given $R = \{(1,-1)\ (1,0)\ (2,-1)\ (3,0)\}$



∴ It is not a function because 1 of A is related to both −1 and 0 of B.

(ii) $A = \{1, 2, 3\}$ and $B = \{-1, 0\}$

$S = \{(1,-1)\ (2,-1)\ (3,0)\}$



It is a function.
Because every element of A has unique element in B

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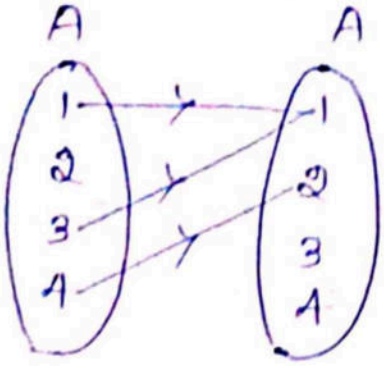
(iii) $A = \{1, 2, 3, 4\}$, $g = \{(3,1)\ (4,2)\ (1,1)\}$

Soln:-

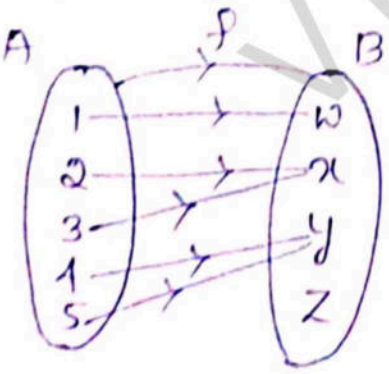


It is not a function because 2 is not related to any element of A (co-domain).

(2) For $A = \{1, 2, 3, 4, 5\}$ and $B = \{w, x, y, z\}$ function f from A to B is defined by $f = \{(1, w)\ (2, x)\ (4, y)\ (5, y)\ (3, x)\}$. Find the images of the following

(i) $A_1 = \{1, 2, 3\}$ (ii) $A_2 = \{2, 3, 4, 5\}$

Soln:-



(i) $A_1 = \{1, 2, 3\}$

The image, $f(A_1) = \{w, x\}$

(ii) $A_2 = \{2, 3, 4, 5\}$

$f(A_2) = \{x, y\}$

VTUJADDA

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(3) Let f from R to R defined by

$f \to$ image
$f^{-1} \to$ preimage

$$f(x) = \begin{cases} 3x-5 & \text{for } x > 0 \\ -3x+1 & \text{for } x \leq 0 \end{cases}$$

(i) Determine the $f(0)$, $f(-1)$, $f(5/3)$, $f(-5/3)$

(ii) Find $f^{-1}(0)$, $f^{-1}(1)$, $f^{-1}(-1)$, $f^{-1}(3)$, $f^{-1}(-6)$

(iii) Find $f^{-1}([-5,5])$ and $f^{-1}([-6,5])$

Soln:- (i) $f(x) = -3x+1$, for $x \leq 0$

$f(x) = 3x-5$, for $x > 0$

$f(0) = -3(0)+1 = 1$

$f(-1) = -3(-1)+1 = 5 \quad (x=-1<0)$

$f(5/3) = 3\left(\frac{5}{3}\right)-5 = 0 \quad (x = 5/3 > 0)$

$f(-5/3) = -3(-5/3)+1 = 6 \quad (x = -5/3 < 0)$

(ii) $f(a) = b \Rightarrow a = f^{-1}(b)$

Let $f^{-1}(0) = x$

$\Rightarrow f(x) = 0$

$3x-5 = 0$

$\boxed{x = 5/3}$

$-3x+1 = 0$

$\boxed{x = 1/3}$

$f^{-1}(0) \neq 1/3 \quad (\because 1/3 < 0)$

$\therefore f^{-1}(0) = \frac{5}{3} \quad (\because \frac{5}{3} > 0)$

To find, $f^{-1}(1)$, Let $f^{-1}(1) = x$

$\Rightarrow f(x) = 1$

VTUADDA

$3x-5=1$; $-3x+1=1$

$\boxed{x=2}$ $\boxed{x=0}$

$f^{-1}(1)=2 \quad (2>0)$

$f^{-1}(1)=0 \quad (0<0)$

$\therefore f^{-1}(1)=\{2,0\}$

To find $f^{-1}(-1)$

$f^{-1}(-1)=x \Rightarrow f(x)=-1$

$3x-5=-1$

$3x=4$

$\boxed{x=4/3}$. $\frac{4}{3}>0$

$-3x+1=-1$

$\Rightarrow -3x=-2$

$\boxed{x=2/3}$ $\frac{2}{3}\not<0$

$\therefore f^{-1}(-1)=\frac{4}{3}$ //

To find $f^{-1}(3)$

$f^{-1}(3)=x \Rightarrow f(x)=3$

$3x-5=3$

$3x=8$

$x=\frac{8}{3}$. $\frac{8}{3}>0$

$-3x+1=3$

$-3x=2$

$x=\frac{-2}{3}$, $\frac{-2}{3}<0$

$\therefore f^{-1}(3)=\frac{8}{3}$, $f^{-1}(3)=-2/3$

$\therefore f^{-1}(3)=\left\{\frac{-2}{3},\frac{8}{3}\right\}$ //

VTUADDA

To find $f^{-1}(-6)$

$f^{-1}(-6) = x \Rightarrow f(x) = -6$

$f(x) = 3x - 5 = -6$

$3x = -1$

$x = -1/3$, $\frac{-1}{3} \not> 0$

$-3x + 1 = -6$

$-3x = -7$

$x = 7/3$, $7/3 \not< 0$

$\therefore f^{-1}(-6) = \phi.$

(iii) $f^{-1}([-5, 5]) = x$

$f(x) = [-5, 5]$

$\Rightarrow -5 \leq f(x) \leq 5$

case (i) $-5 \leq 3x - 5 \leq 5$

$-5+5 \leq 3x - 5 + 5 \leq 5+5$

$0 \leq 3x \leq 10$

$\Rightarrow \frac{0}{3} \leq x \leq \frac{10}{3}$

case (ii) $-5 \leq f(x) \leq 5$

$-5 \leq -3x + 1 \leq 5$

$\Rightarrow -5-1 \leq -3x + \cancel{1} - \cancel{1} \leq 5-1$

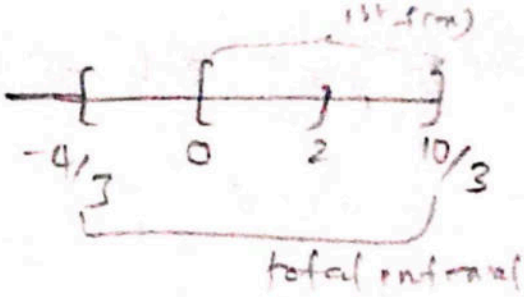
$\Rightarrow -6 \leq -3x \leq 4$

$\Rightarrow 6 \geq 3x \geq -4$ $\div$ by 3

$\Rightarrow 2 \geq x \geq -4/3$ (or) $-4/3 \leq x \leq 2$

VTUADDA

The interval is $\left[-\frac{4}{3}, \frac{10}{3}\right]$ //



<u>To find</u> $f^{-1}([-6,5]) = x$

$$f(x) = [-6, 5]$$

$$\Rightarrow -6 \leq f(x) \leq 5$$

<u>case(i)</u> $-6 \leq 3x-5 \leq 5$

$$-6+5 \leq 3x \leq 5+5$$

$$-1 \leq 3x \leq 10$$

$$\boxed{-\frac{1}{3} \leq x \leq \frac{10}{3}}$$

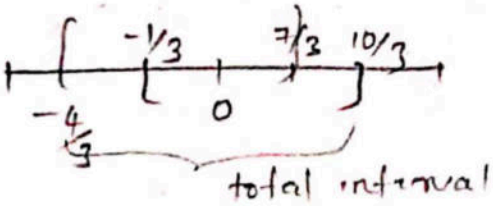
<u>case(ii)</u> $-6 \leq -3x+1 \leq 5$

$$-6-1 \leq -3x \leq 5-1$$

$$-7 \leq -3x \leq 4$$

$$\Rightarrow 7 \geq 3x \geq -4$$

$$\boxed{\frac{7}{3} \geq x \geq -\frac{4}{3}}$$



$\therefore$ The interval is $[-4/3, 10/3]$ //

VTUADDA

(a) If there are 243 functions from A to B and $|B| = 3$, what is $|A|$?

Sol: If A contains 'n' elements and 'B' contains m elements then $f: A \to B$ contains m^n elements

$f: A \to B = 243$, $m = 3$, $n = ?$

But $f: A \to B = m^n$

$243 = 3^n$

$\log(243) = n \log 3$

$\log(243) = n \log 3$ ('ln' in calci)

$\Rightarrow \frac{\log(243)}{\log 3} = n$

$\Rightarrow \boxed{n = 5}$

$\therefore |A| = 5$ //

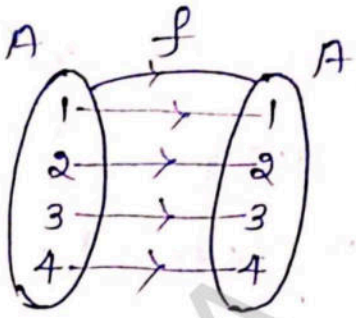
VTUADDA

(*) Types of Functions

(1) Identity functions :-

A function $f: A \to A$ is said to be an identity function, if the images of every element of A is A itself i.e $f(a) = a$ where $a \in A$ and $a \in A$

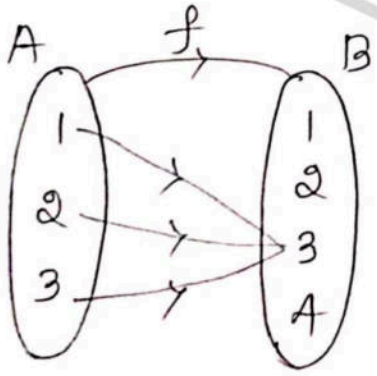
Eg:- $A = \{1, 2, 3, 4\}$



(2) Constant function :-

A func $f: A \to B$ is said to be an constant func if all the elements of A have the same image in B.

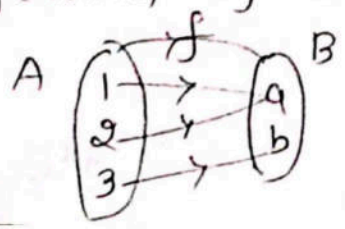
ex:- $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4\}$



(3) Onto function / Subjective function :-

A function $f: A \to B$ is said to be On-to function if every element of B has a pre-image in A.

Ex:- $A = \{1, 2, 3\}$ & $B = \{a, b\}$



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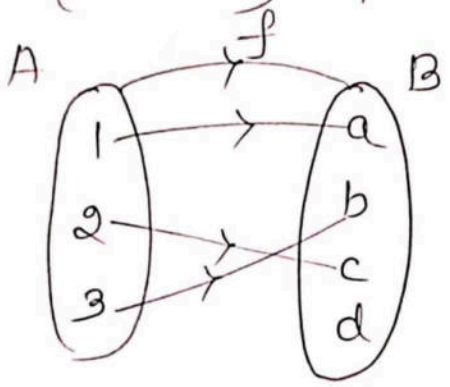
Note: If $|A| < |B|$ then it is not onto.

(4) One-to-one function / Injective function:-

A func $f: A \to B$ is said to be one-to-one func. if every element of A has a unique image in B and every element of $f(a)$ has a unique pre-image in A.

ex:- $A = \{1, 2, 3\}$ & $B = \{a, b, c, d\}$

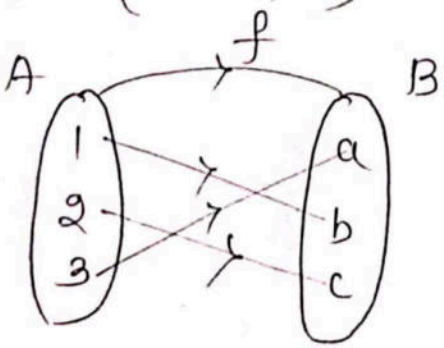


Note:- If $|A| > |B|$ then it is not one-to-one

(5) One-to-one correspondence / Bijective func:

A func $f: A \to B$ is said to be a bijective func if it is both one-one and onto func.

ex:- $A = \{1, 2, 3\}$, $B = \{a, b, c\}$



Note:- If $|A| = |B|$ then only it is a bijective func.

VTUADDA

problems

(1) Find the nature of the following func

$A = \{1, 2, 3\}$

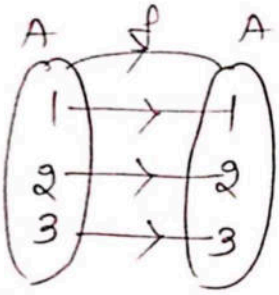
(i) $f = \{(1,1)\ (2,2)\ (3,3)\}$

(ii) $g = \{(1,2)\ (2,2)\ (3,2)\}$

(iii) $h = \{(1,2)\ (2,2)\ (3,1)\}$

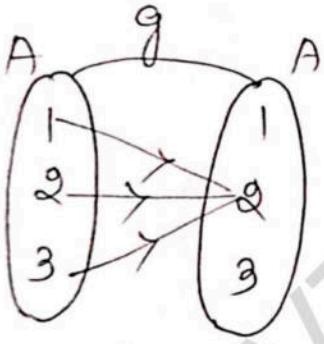
(iv) $p = \{(1,2)(2,3)(3,1)\}$

Soln:- (i)



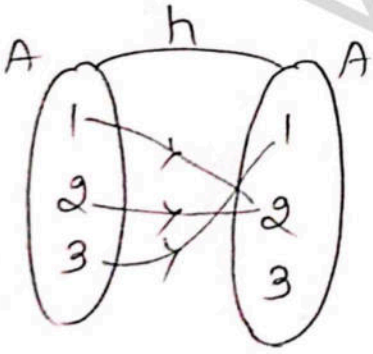
f is identity func.

(ii)



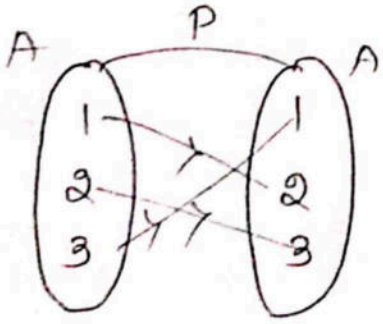
∴ It's a constant func

(iii)



∴ It's a neither one-one nor onto func.

(iv)



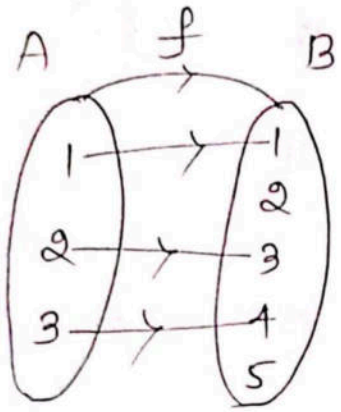
∴ p is bijective func.

VTUADDA

(2) $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4, 5\}$. Find whether it is one-one (or) onto (i) $f = \{(1,1)\ (2,3)\ (3,4)\}$

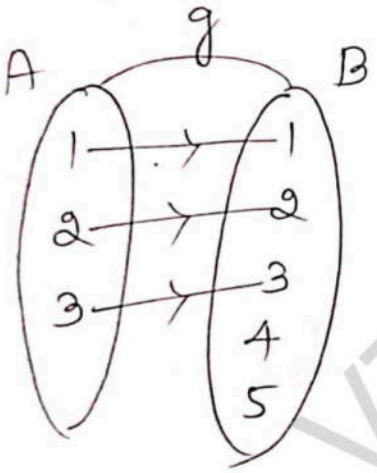
(ii) $g = \{(1,1)\ (2,2)\ (3,3)\}$

Soln:- (i)



$\therefore$ f is one-one function but not onto.

(ii)



$\therefore$ g is one-one function but not onto.

(3) The function $f: R \to R$ and $g: R \to R$ are defined by $f(x) = 3x + 7$ for all $x \in R$ and $g(x) = x(x^3 - 1)$ for all $x \in R$. Verify that f is one-one but g is not.

Soln:- $f(x) = 3x + 7$, $x \in R$

Let us take x_1 & $x_2 \in R$

consider $f(x_1) = f(x_2)$

$$3x_1 + \cancel{7} = 3x_2 + \cancel{7}$$

$$\cancel{3}x_1 = \cancel{3}x_2 \Rightarrow x_1 = x_2$$

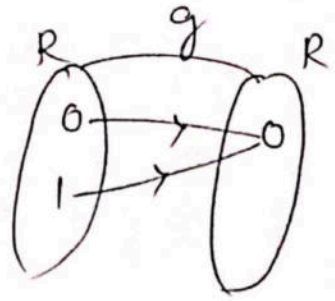
$\therefore$ f is one-one function.

VTUADDA

$g(x) = x(x^3 - 1)$, $x \in R$

take x_1 & $x_2 \in R$ (Not possible)

ie



So, let $x = 0$, $g(0) = 0(0-1) = 0$

$x = 1$, $g(1) = 1(1^3 - 1) = 0$

$\therefore$ 'g' is not one-one function.

(4) Let $f : Z \rightarrow Z$ defined by $f(a) = a + 1$ for $a \in Z$

Find whether f is one-one (or) onto.

<u>Soln:-</u> $f(a) = a + 1$, $a \in Z$

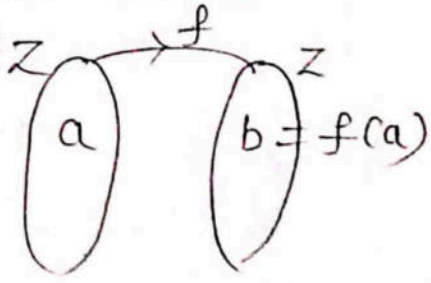
take a_1 & $a_2 \in Z$

consider. $f(a_1) = f(a_2)$

$a_1 + \cancel{1} = a_2 + \cancel{1}$

$\Rightarrow a_1 = a_2$ $\therefore f$ is one-one

Now, For proving onto func, consider $b \in Z$



$f(a) = b$

$a + 1 = b$

$a = b - 1$

$\Rightarrow \because b \in Z \Rightarrow b - 1 \overset{\text{also}}{\in} Z$

$\therefore f$ is onto.

VTUADDA

(5) Let $A = R$, $B = \{x / x \text{ is real } \& \ x \geq 0\}$. If the function $f: A \to B$ defined by $f(a) = a^2$

Soln:- $f(a) = a^2$

$a_1, a_2 \in R$

consider, $f(a_1) = f(a_2)$

$$a_1^2 = a_2^2$$

$$a_1 = \pm\sqrt{a_2^2}$$

$$\therefore a_1 = \pm a_2$$

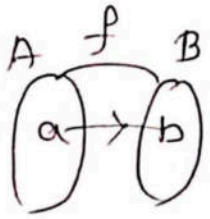
i.e $a_1 = a_2$ (or) $a_1 = -a_2$

$\therefore f$ is not one-one

Now, take $b \in B$, $f(a) = b$

$$a^2 = b$$

$$\Rightarrow a = \pm\sqrt{b}$$



($\pm\sqrt{b} \in A$ because $A = R$)

$\therefore f$ is onto.

Note:-

(1) $|A| = m$, $|B| = n$, f is one-one from A to B

$$p(n, m) = \frac{n!}{(n-m)!}$$

(2) $|A| = m$, $|B| = n$, f is onto func from A to B

$$p(n, m) = \sum_{k=0}^{n} (-1)^k \, {}^nC_{n-k} \, (n-k)^m$$

VTUADDA

(6) There 60 one-one function A to B and $|A| = 3$ What is $|B| = ?$

Soln:- $p(n, m) = \dfrac{n!}{(n-m)!}$

$$60 = \frac{n!}{(n-3)!}$$

$$60 = \frac{n(n-1)(n-2)\cancel{(n-3)!}}{\cancel{(n-3)!}}$$

$$\Rightarrow 60 = n(n-1)(n-2) \qquad [60 = 5 \times 4 \times 3]$$

$$\therefore \boxed{n = 5}$$

Hence $|B| = 5$ //

(7) Let $A = \{1, 2, 3, 4, 5, 6, 7\}$ and $B = \{w, x, y, z\}$. Find the no of onto functions from A to B.

Soln:- $|A| = 7 = m$; $|B| = n = 4$

$$p(n, m) = \sum_{K=0}^{n} (-1)^K \, {}^nC_{n-K} \, (n-K)^m$$

$$p(4, 7) = \sum_{K=0}^{4} (-1)^K \, {}^4C_{4-K} \, (4-K)^m$$

$K = 0$, $p(4, 7) = {}^4C_4 \, 4^7 = 16384$

$K = 1$, $p(4, 7) = -{}^4C_3 \, 3^7 = -4 \times 2187 = -8748.$

$K = 2$, $p(4, 7) = {}^4C_2 \, 2^7 = 6 \times 128 = 768$

$K = 3$, $p(4, 7) = -{}^4C_1 \, 1^7 = -4$

VTUADDA

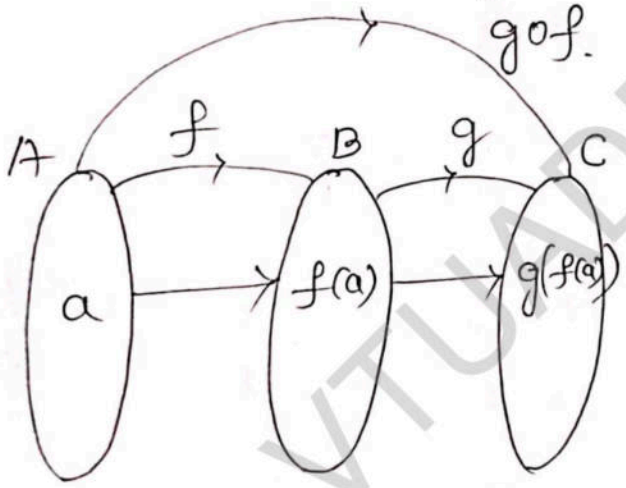
$k=4$, $p(4,7) = (-1)^4\, {}^4C_0\, (0) = 0.$

$\therefore\ p(4,7) = 16384 - 8748 + 768 - 4 + 0$

$= 8400$ //.

(*) Composition of functions

Consider 3 non-empty sets A, B, C and the functions $f: A \to B$ and $g: B \to C$. The composition of these 2 functions is defined as the function $gof: A \to C$ with $gof(a) = g[f(a)]$ where $a \in A$



Note:-

1. $f: A \to A$ then $fof = f^2$

   $fof^2 = f^3$

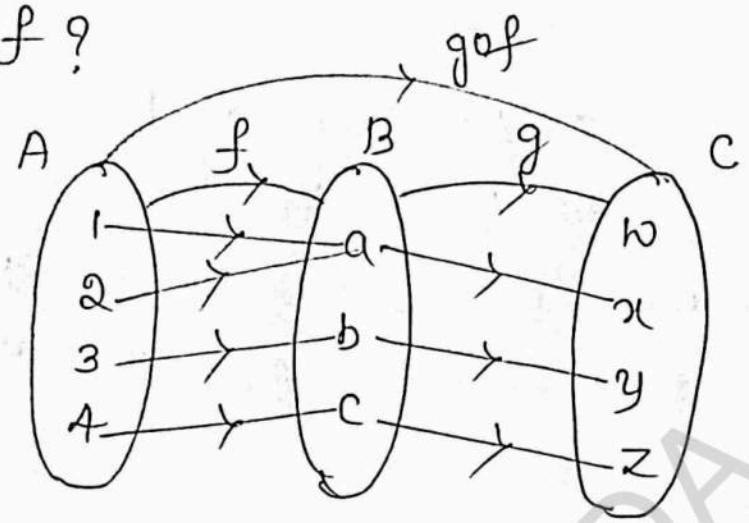
2. $gof(x) = g(f(x))$
3. $fog(x) = f(g(x))$

VTUADDA

problems

(1) Let $A=\{1,2,3,4\}$, $B=\{a,b,c\}$ and $C=\{w,x,y,z\}$ with $f: A \to B$ and $g: B \to C$ given by $f=\{(1,a)(2,a)(3,b)(4,c)\}$ and $g=\{(a,x)(b,y)(c,z)\}$ find gof?

Soln:-



$\therefore gof = \{(1,x)(2,x)(3,y)(4,z)\}$ //

NP (2) consider the functions f and g defined by $f(x)=x^3$ and $g(x)=x^2+1$, $\forall x \in R$. Find gof, fog, f^2 and g^2. And also show that $fog \neq gof$

Soln:- WKT $gof(x) = g(f(x))$

To find gof

$$gof(x) = g(f(x))$$
$$= g(x^3)$$
$$= (x^3)^2+1$$
$$gof = x^6+1 \text{ //}$$

To find fog

$$fog(x) = f(g(x))$$
$$= f(x^2+1)$$
$$fog = (x^2+1)^3$$

$\therefore gof(x) \neq fog(x)$.

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To find f^2

$$f \circ f(x) = f(f(x))$$
$$= f(x^3)$$
$$= (x^3)^3$$
$$f \circ f(x) = x^9$$

To find g^2

$$g \circ g(x) = g(g(x))$$
$$= g(x^2+1)$$
$$g \circ g = (x^2+1)^2+1$$

Ex (3) Let f and g be the functions from R to R defined by $f(x) = ax+b$ and $g(x) = 1-x+x^2$, if $g \circ f(x) = 9x^2-9x+3$. Find a and b.

Soln:- $g \circ f(x) = 9x^2-9x+3$

$$g(f(x)) = 9x^2-9x+3$$
$$g(ax+b) = 9x^2-9x+3$$
$$1-(ax+b)+(ax+b)^2 = 9x^2-9x+3$$
$$1-ax-b+a^2x^2+b^2+2abx = 9x^2-9x+3$$
$$a^2x^2+(2ab-a)x+(1-b+b^2) = 9x^2-9x+3$$

Equating the co-efficients of x^2, x & constants.

x^2: $a^2 = 9 \Rightarrow a = \pm 3$

x: $2ab-a = -9$, when $a=3$, $6b-3=-9$

$\boxed{b=-1}$

When $a=-3$, $-6b+3=-9$

$\boxed{b=2}$

VTUADDA

(*) Theorem 1

Let $f: A \to B$ and $g: B \to C$ be any two functions. Then the following are true

(i) If f and g are one-one, so gof is also one-one

(ii) If gof is 1-1 then f is 1-1

(iii) If f and g are onto so gof is also onto.

(iv) If gof is onto then g is onto.

proof:- If $f: A \to B$ and $g: B \to C$ then

$gof: A \to C$.

(i) To prove gof is one-one ($gof: A \to C$

consider, $a_1, a_2 \in A$

$$gof(a_1) = gof(a_2)$$

$$g(f(a_1)) = g(f(a_2)) \quad (\because g \text{ is } 1\text{-}1)$$

$$f(a_1) = f(a_2) \quad (\because f \text{ is } 1\text{-}1)$$

$$\Rightarrow a_1 = a_2$$

$\therefore gof$ is one-one.

(ii) take, $a_1, a_2 \in A$ then $f(a_1), f(a_2) \in B$ and

$$f(a_1) = f(a_2)$$

take 'g' on. b.s

$$g(f(a_1)) = g(f(a_2)) \quad (\because gof \text{ is } 1\text{-}1)$$

$$\Rightarrow gof(a_1) = gof(a_2) \Rightarrow a_1 = a_2$$

$\therefore f$ is one-one

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(iii) Take $a \in A$, Since f is onto $\Rightarrow f(a) = b$

and since g is onto $\Rightarrow c = g(b)$

consequently, $gof(a) = g[f(a)] = g(b) = c$

$\therefore$ gof is onto.

(iv) Since gof is onto.

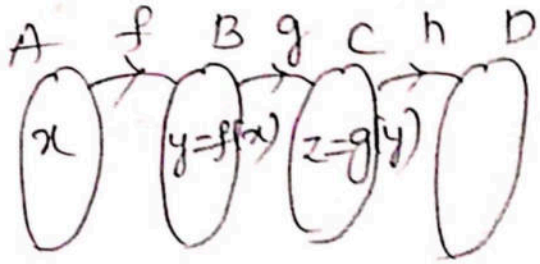
$$gof(a) = g[f(a)]$$
$$= g(b)$$
$$= c \quad , \quad c \in C$$

$\therefore$ g is onto//.

(*) Theorem 2:

Let $f: A \to B$, $g: B \to C$ and $h: C \to D$ be three functions then prove that

$(hog)of = ho(gof)$



proof:- consider LHS

$\{[hog]of\}(x)$, $x \in A$

$$\Rightarrow (hog)of(x) = (hog)(f(x))$$
$$= hog(y) \quad (\because y = f(x))$$
$$= h(g(y)) \quad (\because z = g(y))$$
$$= h(z)$$
$$= LHS$$

Now,

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consider RHS,

$$[ho(gof)](x) = h(gof(x))$$
$$= h(g(f(x)))$$
$$= h(g(y))$$
$$= h(z)$$
$$= RHS$$

$\therefore$ LHS = RHS

Hence proved.

Qn (4) Let f, g, h be the functions from Z to Z defined by $f(x) = x-1$, $g(x) = 3x$, $h(x) = 0$, if x is even and $h(x) = 1$, if x is odd Determine $[fo(goh)](x)$ and $[(fog)oh](x)$ and verify that $fo(goh) = (fog)oh$.

Soln:- $[fo(goh)](x) = f[goh(x)]$

$$= f[g(h(x))]$$
$$= f(3h(x))$$
$$= 3h(x) - 1$$
$$= 3h(x) - 1$$

$g(x) = 3x$ & $h(x) = 0$

$$\Rightarrow \begin{cases} 3(0)-1 = -1, & \text{if } x = \text{even} \\ 3(1)-1 = 2, & \text{if } x = \text{odd} \end{cases}$$

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$$[(f \circ g) \circ h](x) = f \circ g(h(x))$$
$$= f(g(h(x)))$$
$$= f(3(h(x)))$$
$$= 3h(x) - 1$$
$$\Rightarrow \begin{cases} -1, & \text{if } x \text{ is even} \\ 2, & \text{if } x \text{ is odd} \end{cases}$$

$$\therefore f \circ (g \circ h) = (f \circ g) \circ h$$

VTUADDA